Design Method

4-bit subtractor

- Idea: A - B = A + NOT(B) + 1
- Built from four 1-bit full adders, chained by carry (Cout to next Cin)
- NOT gates invert every bit of B; first Cin is the constant 1
- Three hex displays show A, B and the result
- Verified by simulation: 6 - 2 = 4, 5 - 7 = E, 9 - 3 = 6

7-segment decoder

- Input: 4 bits D3..D0, outputs: segments Sa..Sg
- Step 1: truth table for 0-9 and A-F
- Step 2: K-maps for Sa and Sb
- Step 3: minimized Boolean equations (sum of products)
- Step 4: circuit with NOT, AND, OR gates in Digital

Truth Tables

1-bit full adder

A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Decoder: segments a and b (1 = on)

Digit	D3 D2 D1 D0	Sa	Sb
0	0000	1	1
1	0001	0	1
2	0010	1	1
3	0011	1	1
4	0100	0	1
5	0101	1	0
6	0110	1	0
7	0111	1	1
8	1000	1	1
9	1001	1	1
A	1010	1	1
b	1011	0	0
C	1100	1	0
d	1101	0	1
E	1110	1	0
F	1111	1	0

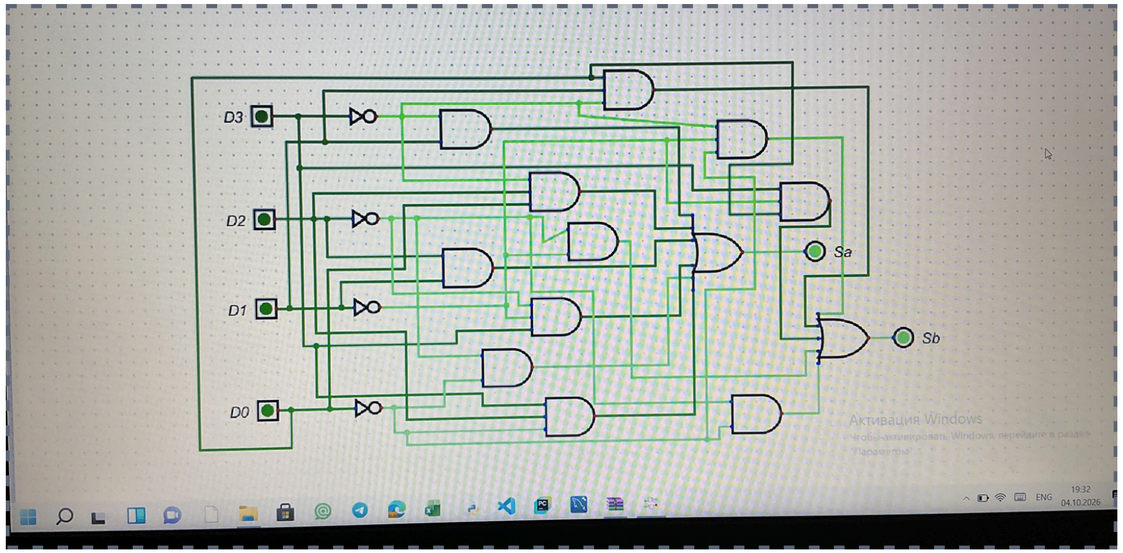
Boolean Expressions

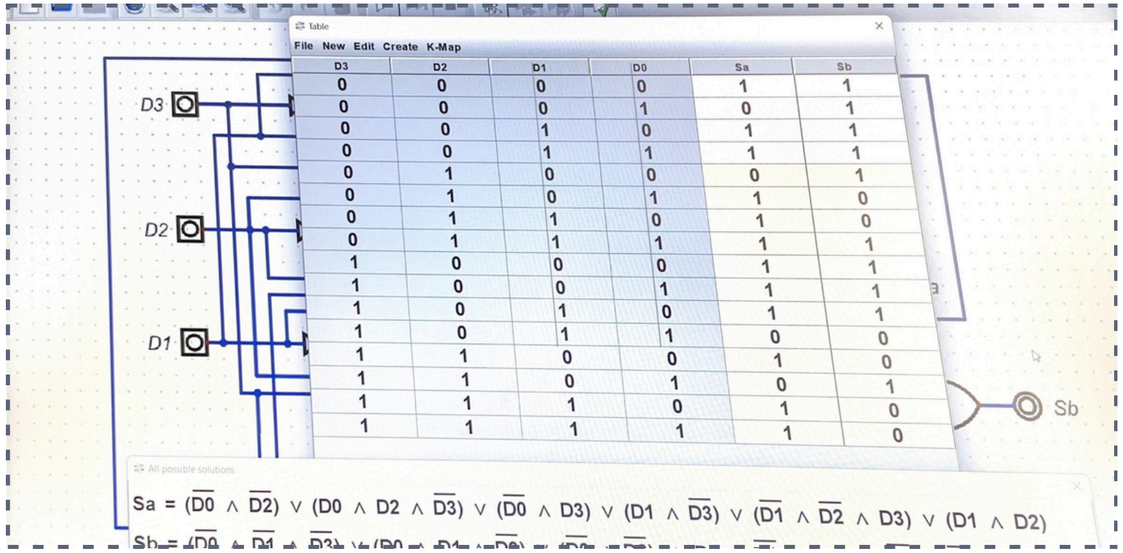
Full adder S = A xor B xor Cin Cout = A·B + Cin·(A xor B)

Subtractor A - B = A + NOT(B) + 1 (Cin of first adder = 1)

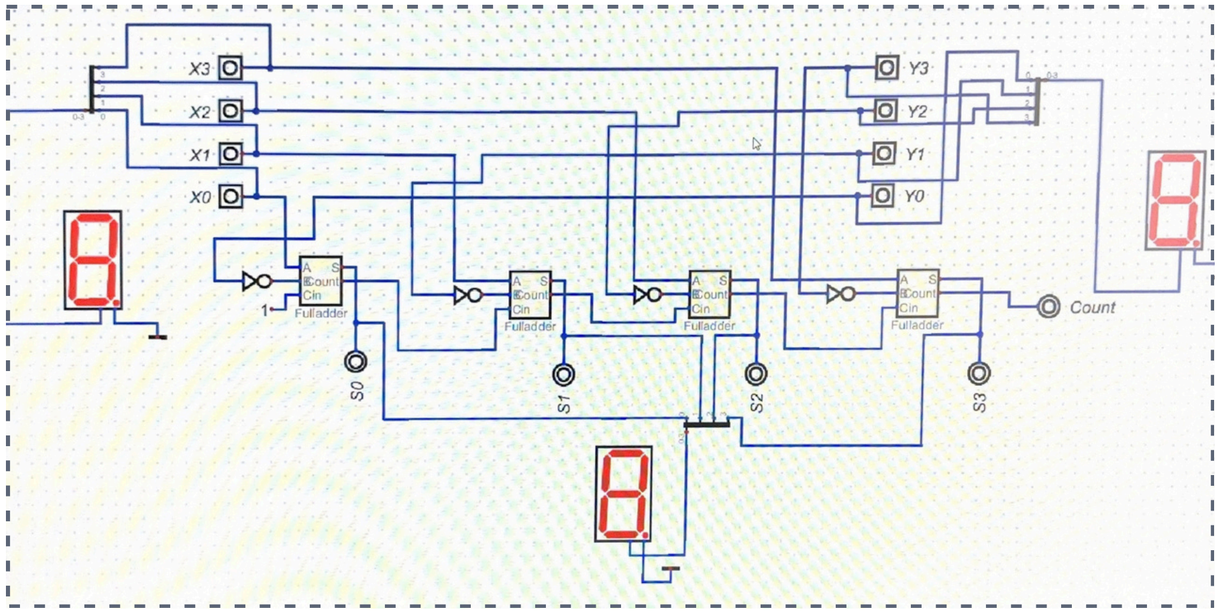
Sa D2'D0' + D3'D1 + D2D1 + D3'D2D0 + D3D2'D1' + D3D2D0'

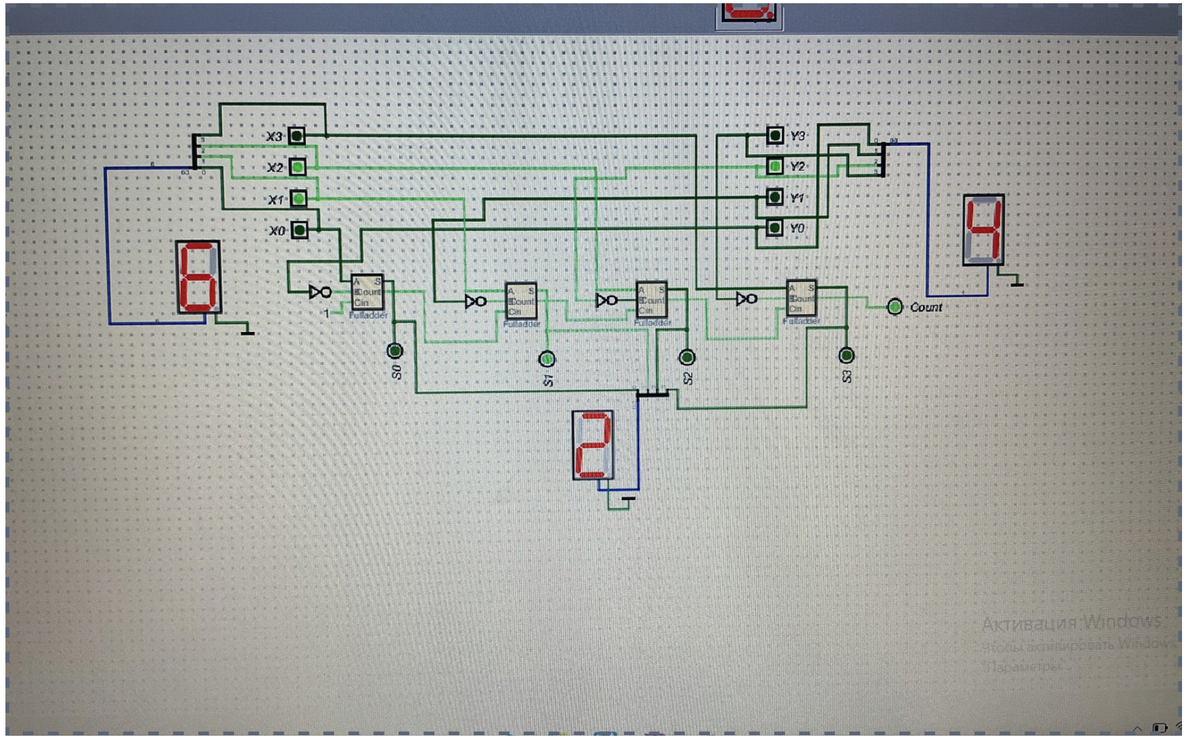
Sb D2'D1' + D2'D0' + D3'D1D0 + D3'D1'D0' + D3D1'D0





Logic Circuit Design





Simulation checks: 6 - 2 = 4 | Decoder matches truth table for all 16 inputs